

Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 2 – Basic Hands-ON and Method Exploring

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I. WORK DONE THIS WEEK:-

This week we have primarily focused on the understanding of tumor presence in brain. We have taken some scans from the dataset and try to localte some tumor and identify it's properties (characteristics). We have also attempted a python script to natively open the MRI scan(.nii.gz) files. There is one software named 'MRICroGL' to view MRI scans with simple UI.

We have gone through some papers based on frequency domain method for brain tumor along with an annotated brain tumor dataset paper. These papers can provide a broad perspective for the upcoming project components.

This week, our primary focus was on developing a fundamental understanding of brain tumor presence and its visual characteristics in MRI scans. We worked with a subset of MRI scans from the available dataset and examined them to identify regions that may have a tumor tissue present. Along with identifying the possible tumor regions, we also began observing their characteristics, such as differences in intensity, shape, size, location, and appearance relative to the surrounding brain tissue. This initial exploration helped us understand the challenges involved in identifying tumor regions from normal brain structures.

As part of the preprocessing and visualization work, we also explored methods for natively opening and examining .nii.gz (NIfTI) MRI files using Python. The results are shown below in Fig. 1The objective was to understand the underlying structure of the medical imaging data and writtea Python script that can load, process, and analyze the scans without depending entirely on external visualization software. This is

an important step toward automating the later stages of the project.

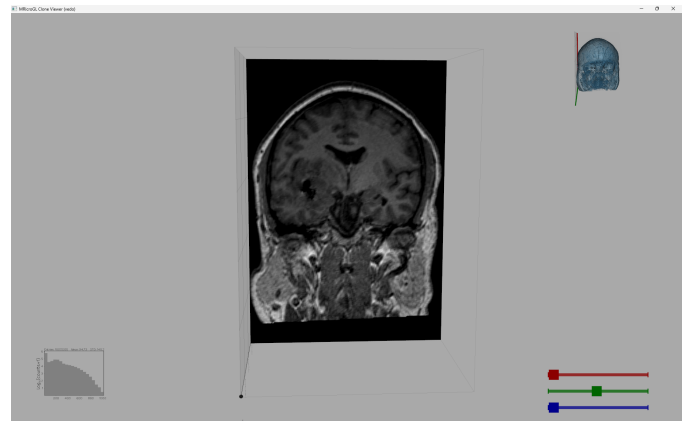


Fig. 1. Python script result

In addition to the Python-based approach, we explored MRICroGL, a dedicated medical-image visualization software that provides a relatively simple interface for viewing and navigating MRI scans as shown in Fig. 2. Using MRICroGL allowed us to inspect the scans more conveniently and develop a better understanding of the spatial structure and visual appearance of brain MRI data. It also provided a useful reference for validating the results obtained through our Python-based visualization approach.

As we mentioned above, we have manually try to examine some dataset so we found some abnormalities in boundaries of brain tumor. As shown in Fig. 3, we can observe a rough patch in the image on the boundaries of tumor.

We also started reviewing research literature related to brain

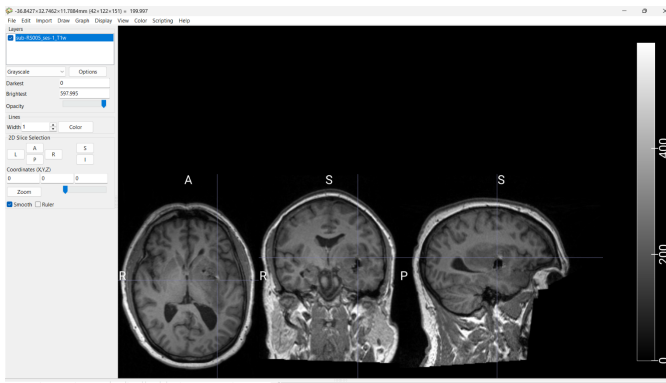


Fig. 2. MRIcroGL UI

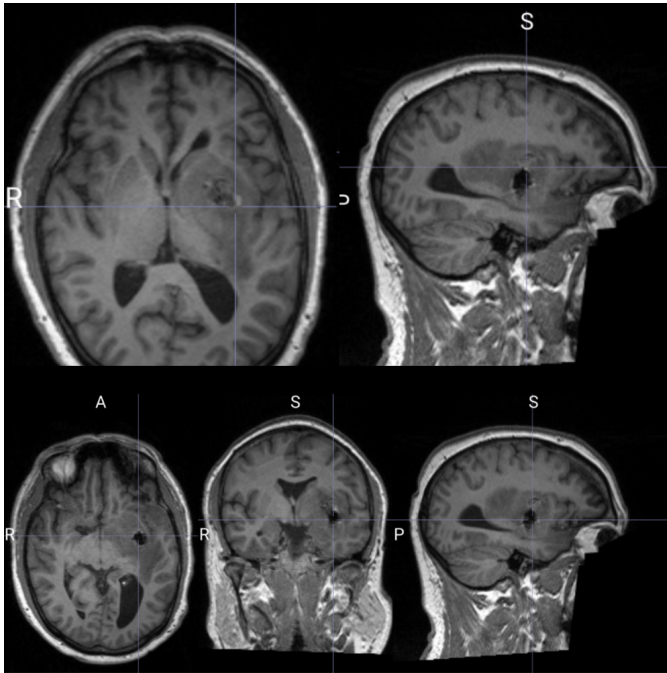


Fig. 3. Visual findings of brain tumor

tumor detection and analysis using frequency-domain methods. These papers provided insights into how transformations such as the Fourier Transform and frequency-domain filtering can be used to extract useful information from MRI images. Understanding these approaches will help us evaluate whether frequency-domain features can contribute to tumor localization or characterization in the later stages of the project.

Alongside the frequency-domain literature, we reviewed a paper describing an annotated brain tumor dataset. The dataset and its annotation methodology are particularly relevant because reliable ground-truth annotations are necessary for evaluating, and validating a tumor detection or segmentation system. Studying the dataset paper also helped us understand the type of annotations, imaging modalities, and metadata that may be available for our project.

Overall, this week's work established the foundation for

the upcoming project components. We now have a better understanding of the MRI data format, basic tumor appearance in brain scans, available visualization tools, and relevant research directions. The literature review has also helped us identify possible approaches for combining image-domain and frequency-domain information. In the upcoming stages, we plan to focus on systematic preprocessing of the MRI scans, tumor localization/segmentation, extraction of relevant image characteristics, and evaluation of frequency-domain techniques on the selected dataset.

Dataset Taken from:-

<https://openneuro.org/datasets/ds005003/versions/2.0.0>

II. WORK TO BE DONE NEXT WEEK:-

The upcoming week, from 20th September to 27th September, will consist of mid-terms, so the next submission will be completed on 3rd October.

Task 1: Contacting faculty members Prof. Jayendra Bhalodiya and Prof. Pooja Shah given course faculty permits. Prof. Bhalodiya and Prof. Shah have conducted research and authored papers on identifying and analysing brain tumors. We plan to seek professional advice and help for these faculties for the same. Their insights and knowledge on brain Tumor segmentation will help us better understand brain tumors. Their identification and size estimation.

Task 2: Using image analysis and transformation methods like changing the image domain from space time to frequency, using different types of kernels and performing convolutions to identify patterns among brain segments. With the help of code and image analysis methods, try form a baseline of gray matter and segment patterns among different sections of brain and among different axes. Try to find inconsistencies in this patten and possibly figure out the tumor

III. LITERATURE REVIEW:-

Through the Paper, "Rethinking Brain Tumor Segmentation From the Frequency Domain Perspective", We basically wanted to find a correlation between what we have learnt in class and What needs to be applied in our project. This Paper which was submitted to the conference of IEEE transactions on Medical Imaging. But when we looked into the paper, we saw that it was actually a highly complex paper in the same field, they were working on converting the Spatial Domain 3D MRI Scans into Frequency Domain using methods like Dual-Tree Complex Wavelet Transform (DTCWT) and Nonsubsampled Contourlet Transform (NSCT) of which we did not have much idea, on our understanding they tried to establish a connection by dividing the samples into those of higher frequency and lower frequency, and analysis is done in the according manner.

They have also given a brief on various methods that they have used via which we realised that we have only understood the concept of Single-Modality Models which suffer due to limitations in contextual understanding, for better understanding we have to look into how to apply the work on Multi-Modality Models for a greater understanding of the flow throughout the time series of Tumor Growth, Most Processes

in those felt new and different to us, requiring further study on such a matter.

Throughout the latter half of the Paper, We also saw that they were using Deep Neural Networks to do so, whereas we have explicitly been told to use Classical Imaging Methods, but this has helped us in better understanding on how to actually visualise and understand the Tumor Models more, giving us a greater inference regarding a basic idea that a domain transformation is possible in a Modal Series too, but we have to apply deeper thought and understanding to further do so. We might also require your guidance on this matter. They have used a Laplacian Convolution Method, which felt similar but different too, we might try to understand and use that in our project in the further weeks.

Whereas the Second Paper that we have looked into this week, talks about the Dataset associated with Brain Tumor Segmentation and Classification, this was our attempt at better understanding and going through the dataset we discovered last week, while also trying to have an alternative dataset if any other issues arise.

This Paper has been a boon for us this week, because it helped us with our initial analysis and inferences that we have made this week, This Paper helped us better understand the medical terms and also helped us understand how to truly segregate a tumor from the other brain matter present in its neighbouring locations. Even though we faced a few similar issues like the last paper of truly understanding a few terminologies, even though our group has basically decided to learn while working on our project throughout the weeks. We again would like to reiterate the fact that this paper basically helped us make our own detections on the dataset that we planned to use in the first week.

We have basically tried to expand upon whatever we learnt from the first week, but we still realise the need to go through more readings, because we still lack much inter depth on the project domain. We also realise the need for being extremely technical in this project, because mixing up wordings both in the Brain Tumor Domain or in the Image Processing Domain is going to cause further disturbances, so we are trying to learn as much as we can throughout, because both of these domains are quite the new concepts for us, but we have strong resolve on implementing the ideas that we have learnt in class onto our very own project.

Repo Link:-

https://github.com/Rushin-Mod/DIP_2026_P20_G9

REFERENCES

- [1] M. Shao et al., "Rethinking Brain Tumor Segmentation From the Frequency Domain Perspective," in *IEEE Transactions on Medical Imaging*, vol. 44, no. 11, pp. 4536-4553, Nov. 2025, doi: 10.1109/TMI.2025.3579213.
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